

Multiple Linear regression project

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Dataset Description:

The dataset I used in this project is Boston Housing dataset , obtained from Kaggle website, and the Sample size is **333**, and There are **15** variables in total.

<https://www.kaggle.com/code/tolgahancepel/boston-housing-regression-analysis>.

The Response variable (Y): "medv" (The Median value of owner homes)

Predictor variables (X) : 14 variables, and I chose three variables for they strong relationship with the Response variable, and they are

1: "rm" , Number of rooms in the House.

2: "crim", crime rate.

3: "lstat" , the percentage of low-income population.

ID	crim	zn
Min. : 1	Min. : 0.00632	Min. : 0.00
1st Qu.:123	1st Qu.: 0.07896	1st Qu.: 0.00
Median :244	Median : 0.26169	Median : 0.00
Mean :251	Mean : 3.36034	Mean : 10.69
3rd Qu.:377	3rd Qu.: 3.67822	3rd Qu.: 12.50
Max. :506	Max. :73.53410	Max. :100.00

indus	chas	nox
Min. : 0.74	Min. : 0.00000	Min. : 0.3850
1st Qu.: 5.13	1st Qu.: 0.00000	1st Qu.: 0.4530
Median : 9.90	Median : 0.00000	Median : 0.5380
Mean :11.29	Mean : 0.06006	Mean : 0.5571
3rd Qu.:18.10	3rd Qu.: 0.00000	3rd Qu.: 0.6310
Max. :27.74	Max. : 1.00000	Max. : 0.8710

rm	age	dis
Min. :3.561	Min. : 6.00	Min. : 1.130
1st Qu.:5.884	1st Qu.: 45.40	1st Qu.: 2.122
Median :6.202	Median : 76.70	Median : 3.092
Mean :6.266	Mean : 68.23	Mean : 3.710
3rd Qu.:6.595	3rd Qu.: 93.80	3rd Qu.: 5.117

Max. :8.725 Max. :100.00 Max. :10.710

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Min. :1.000 Min. :188.0 Min. :12.60

1st Qu.: 4.000 1st Qu.:279.0 1st Qu.:17.40

Median : 5.000 Median :330.0 Median :19.00

Mean : 9.634 Mean :409.3 Mean :18.45

3rd Qu.:24.000 3rd Qu.:666.0 3rd Qu.:20.20

Max. :24.000 Max. :711.0 Max. :21.20

black lstat medv

Min. : 3.5 Min. : 1.73 Min. : 5.00

1st Qu.:376.7 1st Qu.: 7.18 1st Qu.:17.40

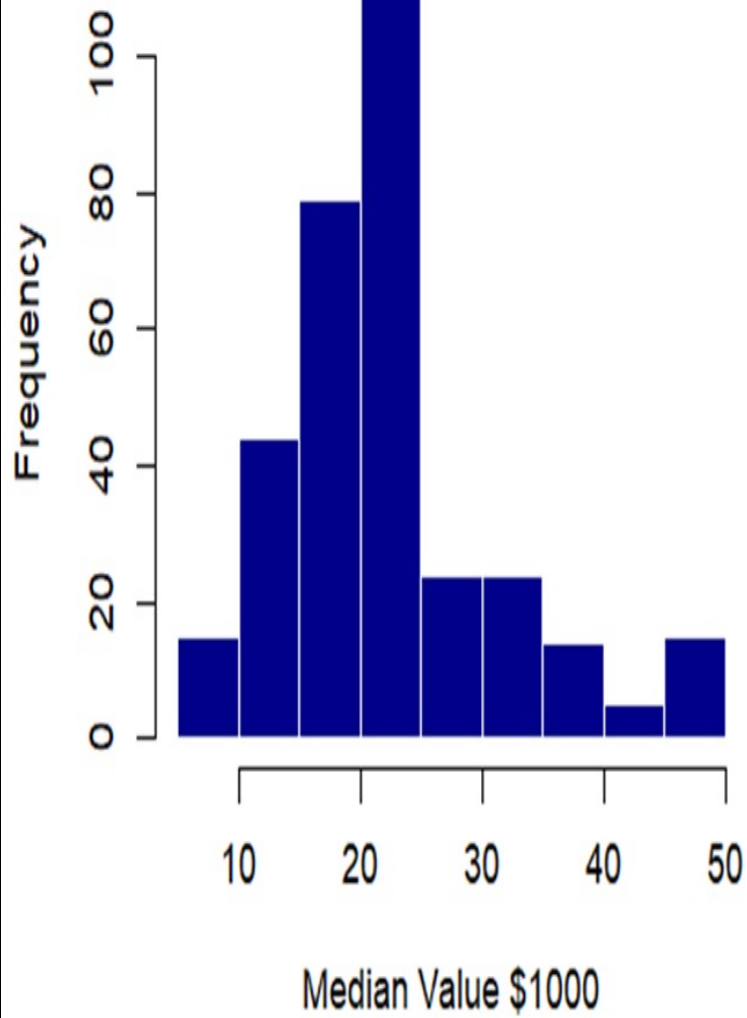
Median :392.1 Median :10.97 Median :21.60

Mean :359.5 Mean :12.52 Mean :22.77

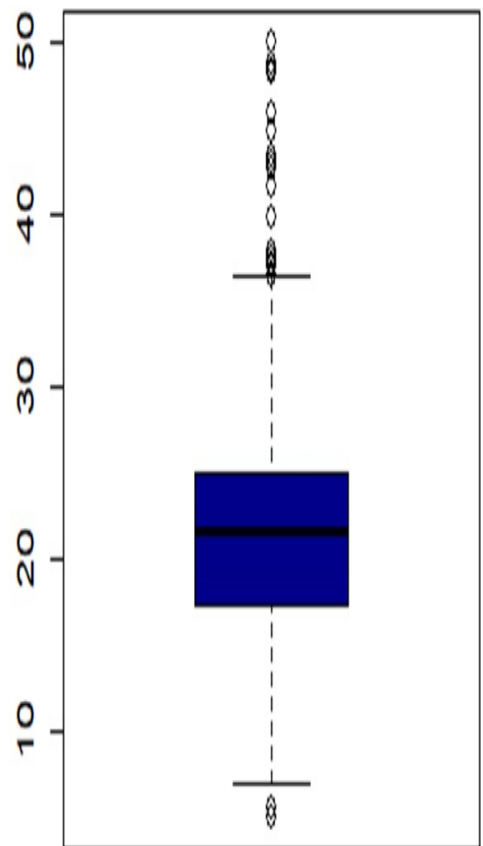
3rd Qu.:396.2 3rd Qu.:16.42 3rd Qu.:25.00

Max. :396.9 Max. :37.97 Max. :50.00

Histogram of Prices (medv)

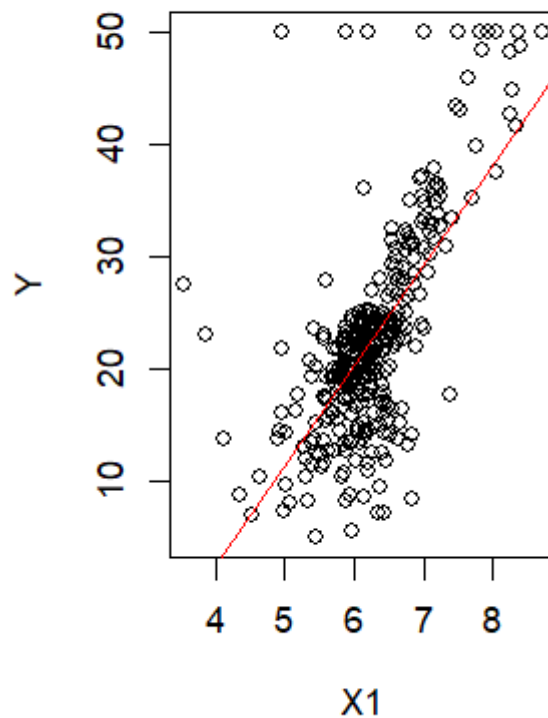


Boxplot of House Prices



Simple Linear Regression:

I built this model with the predictor X1 (which is number of rooms in the house) because it has strong



relationship with response.

Response Variable (Y): (House Price) medv

Predictor Variable (X1): (Number of Rooms) rm

$$\hat{Y} = -33.5367 + 8.9864 X1$$

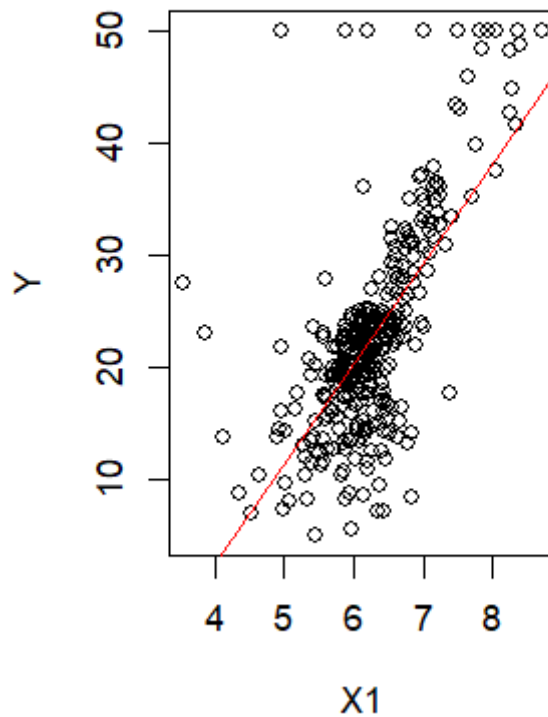
*b*₁; For every unit increase in the number of rooms (X1), the expected house price (Y) increases by **8.98** units.

*b*₀: -33.5367 represents the expected house price and, the house price when the number of rooms =0, which has no meaning here.

R-squared (R^2) and Model Fit:

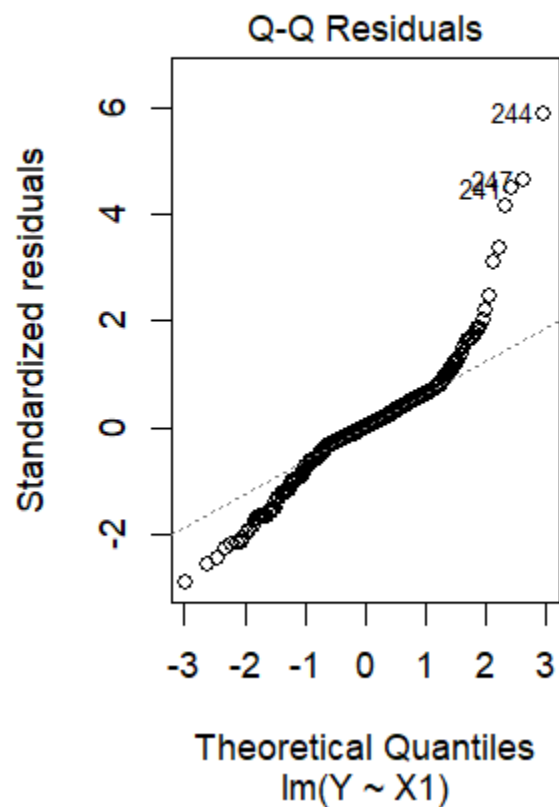
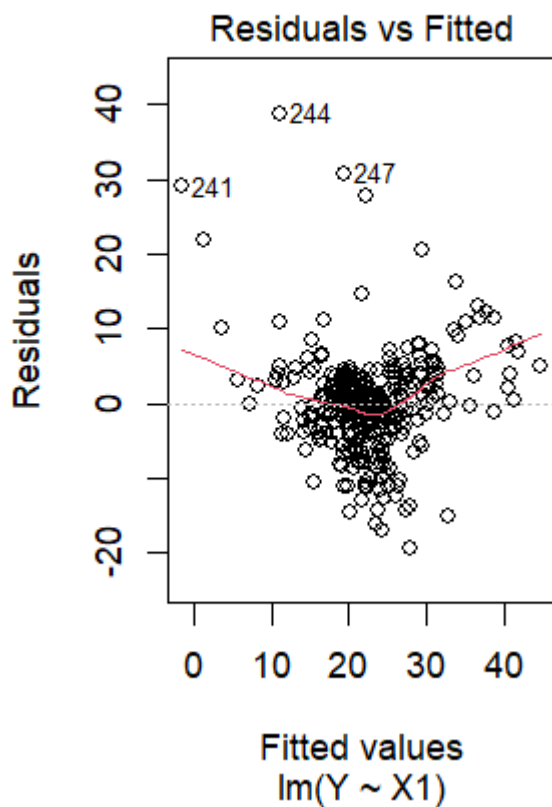
The value of R-squared is **0.4755**, The estimated regression function explains 47.55% of the changes in the house price by using the number of rooms.

Regression Line Plot:



The scatter plot shows a positive linear relationship between the number of rooms and house prices.

Residuals Plots and assumptions:



Linearity: The points are randomly scattered around zero in the 1st plot, and the linearity assumptions satisfy.

Constant Variance: in the 1st plot it shows a funnel shape, and the constant variance assumptions are clearly not satisfied.

Normality: The points in the 2nd plot deviate from the straight line at upper tail, So the normality assumptions is not satisfied.

Inference:

from `summary(model)`, the p-value for the coefficient equal to $< 2e-16$.

statistically significant:

Since the p-value is less than **0.05**, we reject the null hypothesis,

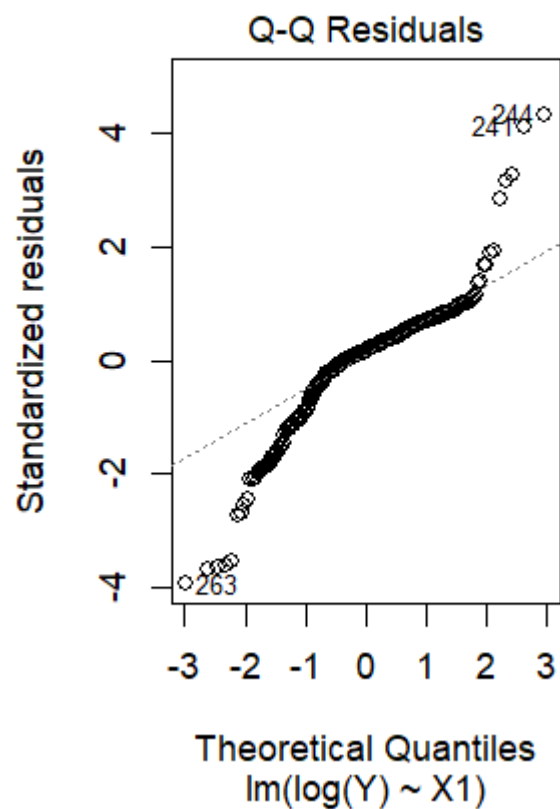
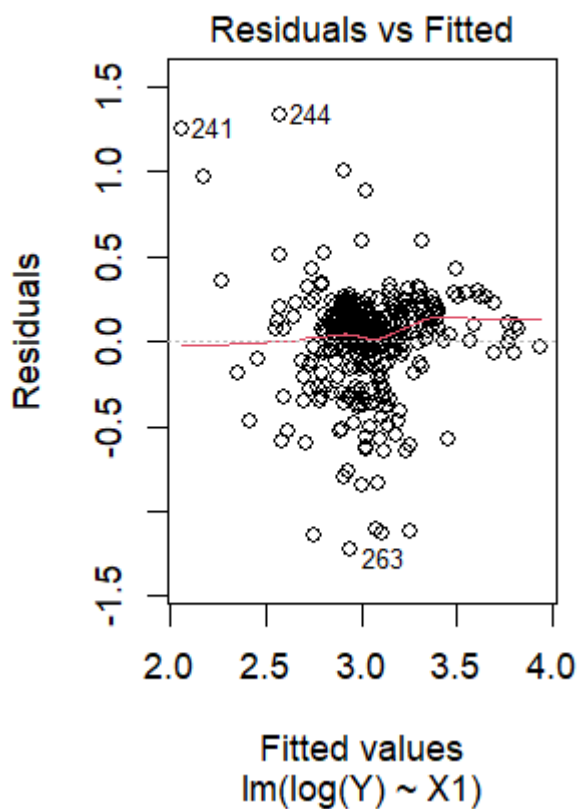
And there is statistically significant relationship between the variables at the 5% level.

Lack Of Fit Test:

The Lack of Fit Test is Not applicable in this case, because there are no exact replicated values "repeated observations" for predictor variable X1.

Since the test applicable in this case, there are no results to interpret.

Transformation:



Based on the residual plots :

Linearity:

Satisfied and randomly scattered.

Constant Variance:

Not satisfied

Normality:

Not satisfied. Therefore, the model assumptions are not reasonable.

We applied a natural log transformation: $\log(Y)$

The transformed model $\text{lm}(\log(Y) \sim X1)$ improves the fit based on the residual plots.

The log transformation stabilized the variance and improved the normality of the residuals.

Multiple Regression:

Response variable (Y): House Price.

Predictor variables (X):

X1: The average room number in house.

X3: The percentage of low-income population.

The fitted model:

$$\hat{Y} = 0.1039 + 4.9285 X1 - 0.6564 X3$$

Interpretation of coefficients:

X1: A one unit increase in the average number of rooms leads to an increase in the house price by 4.9285 units.

X2: A one unit increase in the percentage of low-income population leads to decrease in the house price by 0.6564 units.

Significant Predictors:

Both X1 and X3 p-values < 2e-16.

The multiple model is better than the simple model because it has higher Adjusted R^2 .

Simple model:	Multiple model:
$R^2 = 0.474$	$R^2 = 0.6321$

Communication with a Non-Technical Audience:

This project analyzes real estate data from Boston to help people to make the right decisions by measuring many factors, my project depends on three variables

- 1 :The Average number of rooms in the house.
- 2: The crime rate in the neighborhood.
- 3: The percentage of Low-income people in the Area.

By analyzing these three factors, we can build a good model to predict the price of any house in the neighborhood.

As is known, the more rooms in the house, the higher the price of the house. However, the location matters a lot, and its important as the house itself. If the location of the house is in area with an extremely high crime rate or high poverty, its value will drop surely. By knowing how these three factors work together, people can easily afford a price for any house.

For example, our data shows that adding just one room increases a house price by almost **\$4,900**, on the other side the neighborhood location and condition can decrease the home price as every one-unit increase in the crime rate drops the house price by approximately **\$360**.

The results also shows that an extremely high crime rate and low-income population in the area, have a clear and obviously a negative impact on the house, no matter how big the house is, and it shows for every 1% increase in the low-income population in the area , the house value will decreased by approximately **\$635**.

Therefore, a house's true value is a balance between its size and its environment. This means that checking the neighborhood safety and wealth is important as the house size when figuring out a fair price.